

SUGGESTED BIBLIOGRAPHY

By Spyros Tserkis

December 9, 2025

QUANTUM THEORY (PHYSICS)

- Ballentine - "Quantum Mechanics: A Modern Development"
- Walls, Milburn - "Quantum Optics"
- Serafini - "Quantum Continuous Variables"
- Peres - "Quantum Theory: Concepts and Methods"
- Bohm - "Quantum Mechanics: Foundations and Applications"

QUANTUM THEORY (MATHEMATICS)

- Heinosaari, Ziman - "The Mathematical Language of Quantum Theory"
- Prugovecki - "Quantum Mechanics in Hilbert Space"
- Holevo - "Probabilistic and Statistical Aspects of Quantum Theory"

HISTORY & INTERPRETATION OF QUANTUM MECHANICS

- Jammer - "The Philosophy of Quantum Mechanics"
- Plotnitsky - "Reading Bohr: Physics and Philosophy"

QUANTUM COMPUTING & COMMUNICATION

- Ekert, Hosgood, Kay, Macchiavello - "Introduction to Quantum Information Science"
- Watrous - "Understanding Quantum Information and Computation"
- Riccardo, Motta - "Quantum Information Science"

CLASSICAL & QUANTUM INFORMATION THEORY

- Cover, Thomas - "Elements of Information Theory"
- Holevo - "Quantum Systems, Channels, Information"

LINEAR ALGEBRA & MATRIX ANALYSIS

- Jänich - "Linear Algebra"
- Axler - "Linear Algebra Done Right"
- Halmos - "Finite-Dimensional Vector Spaces"
- Horn, Johnson - "Matrix Analysis"

REAL & FUNCTIONAL ANALYSIS

- Halmos - "Introduction to Hilbert Space and the Theory of Spectral Multiplicity"
- Axler - "Measure, Integration & Real Analysis"

PROBABILITY THEORY & STATISTICS

- Bertsekas, Tsitsiklis - "Introduction to Probability"
- Rényi - "Probability Theory"
- Rényi - "Foundations of Probability"